known from laboratory experiments, too.) Now, another bombardment by a proton changes this carbon nucleus into an ordinary nitrogen nucleus (3). The latter changes, by the absorption of another proton, into an unstable oxygen nucleus (4), which immediately emits a positron and thus becomes a heavy isotope of the first nitrogen nucleus (5). Finally, the entire process is concluded by another absorption of a proton, and the emission of a helium nucleus (in other words, an alpha particle) results once again in an ordinary carbon nucleus.

A ‘balance sheet’ of this series of reactions shows the following picture: First, there is a carbon nucleus, ${}_6C^{12}$, which captures four protons, step by step. At the end, there remains the same ${}_6C^{12}$ nucleus, with a helium nucleus, ${}_2He^4$, plus the two positrons emitted in reactions No. 2 and No. 5. The summary of this ‘balance sheet’ can be put as follows: One helium nucleus and two positrons have been formed from four protons. The net result of the process, therefore, is expressed by the following summary formula:

$$4\,({}_1H^1) \rightarrow {}_2He^4 + 2\,({}_1e^0)$$

The helium nucleus consists of two protons and two neutrons; hence, its charge is two units smaller than the charge of the four protons. This difference is accounted for by the two positrons. Two of the four protons have thus changed into neutrons.

Since the masses of the protons and of the helium nuclei are exactly known, the energy balance of the entire process can be determined on the basis of the last formula. The energy liberated in such a process is 25·5 Mev., or—computed for moles and converted into kilocalories—600,000,000 kilocalories per mole. This is six times as much as in the process mentioned above.

That which takes place here can be expressed as follows: In the interiors of the stars, hydrogen is converted into helium by nuclear ‘combustion’, and this process liberates the vast energies continuously radiated by the sun and stars. It has been said occasionally, in jest, the sun is ‘heated by coal’. But this is not quite correct. The coal—carbon—acts here merely as a catalyst, and is not consumed in the reaction.

This example ought to be sufficient to show that immense stores of energy are liberated in nuclear reactions, if these take place in sufficient quantities of material. Let us further add that there is actually good reason to believe that